

Homework 2: Concise Answers

BFS, DFS, Dijkstra, and A*

BFS1 (p. 1)

Root depth 0; exclude repeated states. Initial board: $\begin{bmatrix} 2 & 8 & 3 \\ 1 & 6 & 4 \\ 7 & \square & 5 \end{bmatrix}$.

Depth d	0	1	2	3	4	5
States N_d	1	3	5	10	14	28

Blank degree: corner 2, edge 3, center 4. Exclude parent reversals:

Depth	Corner blanks	Edge blanks	Center blanks	Total
0	0	1	0	1
1	2	0	1	3
2	0	5	0	5
3	8	0	2	10
4	0	14	0	14
5	20	0	8	28

$$N_2 = 2(2 - 1) + (4 - 1) = 5, N_3 = 5(3 - 1) = 10.$$

Half of depth 5 is examined:

$$N = 1 + 3 + 5 + 10 + 14 + \frac{28}{2} = \boxed{47 \text{ states}}.$$

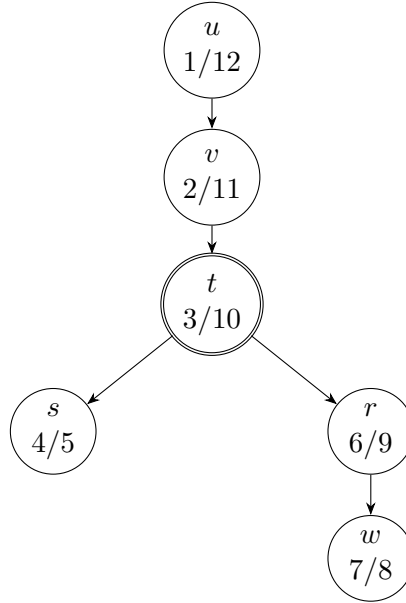
Root and goal included.

DFS1 (p. 2)

Discovery order:

$$u \rightarrow v \rightarrow t \rightarrow s, \quad \text{return to } t, \quad t \rightarrow r \rightarrow w.$$

The tree edges are $(u, v), (v, t), (t, s), (t, r), (r, w)$.



Labels: discovery/finish times; double circle: backtrack node.

Returns:

$$s \rightsquigarrow t, \quad w \rightsquigarrow r \rightsquigarrow t \rightsquigarrow v \rightsquigarrow u.$$

Backtrack node = return point starting a new unvisited branch. $\boxed{bt = t}$.

$E_{\text{back}} = \{(s, u), (w, v)\}$; all remaining edges are tree edges.

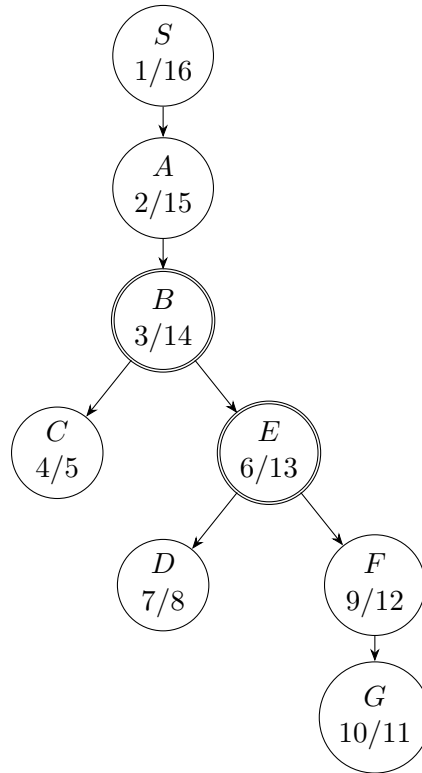
DFS2 (p. 3)

Start S ; alphabetical adjacency; right-hand graph. D is one vertex, visited once.

Discovery order:

S, A, B, C, E, D, F, G .

Labels: discovery/finish times; double circles: backtrack nodes.



$bt = \{B, E\}$:

$C \rightsquigarrow B \rightarrow E, \quad D \rightsquigarrow E \rightarrow F.$

$D \in$ visited when A, S resume: no second D child.

Keep the supplied prefix. Convention for remaining neighbors: ascending rc labels ($rc = 10r + c$). The grid slide gives no fixed directional order.

1. Backtrack to $bf_1 = 41$:

$$(3, 2) \rightsquigarrow (2, 2) \rightsquigarrow (1, 2) \rightsquigarrow (1, 1) \rightsquigarrow (2, 1) \rightsquigarrow (3, 1) \rightsquigarrow (4, 1).$$

Visit:

$$(4, 1) \rightarrow (5, 1) \rightarrow (6, 1) \rightarrow (6, 2).$$

62: dead end (all neighbors visited).

2. Backtrack to $bf_2 = 42$:

$$(6, 2) \rightsquigarrow (6, 1) \rightsquigarrow (5, 1) \rightsquigarrow (4, 1) \rightsquigarrow (4, 2).$$

Visit:

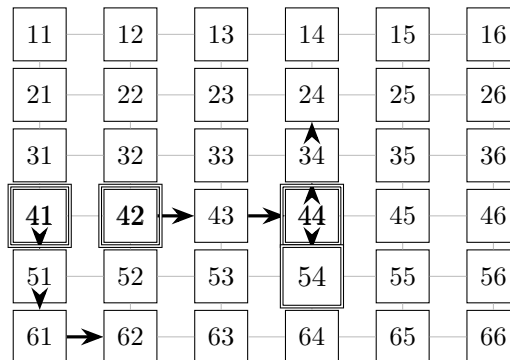
$$(4, 2) \rightarrow (4, 3) \rightarrow (4, 4) \rightarrow (3, 4) \rightarrow (2, 4).$$

34 < 54, so visit 34 first; 24 is a dead end.

3. Backtrack to $bf_3 = 44$:

$$(2, 4) \rightsquigarrow (3, 4) \rightsquigarrow \boxed{bf_3 = (4, 4)}.$$

44 $\rightarrow$ 54: goal; stop. $\boxed{bf_1 = 41, bf_2 = 42, bf_3 = 44.}$



Arrows: new visits. Double boxes: backtrack nodes and goal. Final path:

$$33 \rightarrow 23 \rightarrow 13 \rightarrow 14 \rightarrow 15 \rightarrow 25 \rightarrow 35 \rightarrow 45 \rightarrow 55$$
$$\rightarrow 65 \rightarrow 64 \rightarrow 63 \rightarrow 53 \rightarrow 52 \rightarrow 42 \rightarrow 43 \rightarrow 44 \rightarrow 54.$$

DFS4 (p. 5)

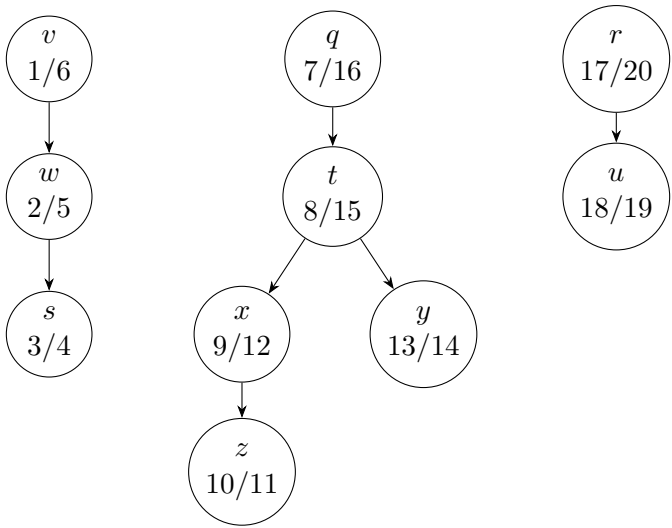
Start v ; remaining roots and adjacency lists alphabetical.

Vertex	Outgoing neighbors	Vertex	Outgoing neighbors
q	s, t, w	r	u, y
s	v	t	x, y
u	y	v	w
w	s	x	z
y	q	z	x

Discovery order:

$v, w, s, q, t, x, z, y, r, u$.

DFS forest (discovery/finish times):



Finish order: $s, w, v, z, x, y, t, q, u, r$. The non-tree edges are:

- Back edges: $s \rightarrow v, z \rightarrow x, y \rightarrow q$.
- Cross edges: $q \rightarrow s, q \rightarrow w, u \rightarrow y, r \rightarrow y$.

Roots: v, q, r .

DFS5 (p. 6)

Reverse DFS finish order:

$p, n, o, s, m, r, y, v, x, w, z, u, q, t$
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Check: $(u, v) \in E \Rightarrow \text{pos}(u) < \text{pos}(v)$.

Vertex	Its outgoing neighbors, all later in the ordering
p	o, s, z
n	o, q, u
o	r, s, v
s	r
m	q, r, x
r	u, y
y	v
v	w, x
w	z
u	t
q	t

t, x, z : no outgoing edges.

Dijkstra1 (p. 7)

Source a ; $w(b, h) = 1$. Initially $d(a) = 0$, all other $d = \infty$.

$d(v) \leftarrow \min\{d(v), d(u) + w(u, v)\}$; settle minimum d (alphabetical ties).

Step	Settled	a	b	h	g	f	c	i	e	d
0	—	0	∞	∞	∞	∞	∞	∞	∞	∞
1	a	0	4	8	∞	∞	∞	∞	∞	∞
2	b	0	4	5	∞	∞	12	∞	∞	∞
3	h	0	4	5	6	∞	12	12	∞	∞
4	g	0	4	5	6	8	12	12	∞	∞
5	f	0	4	5	6	8	12	12	18	22
6	c	0	4	5	6	8	12	12	18	19
7	i	0	4	5	6	8	12	12	18	19
8	e	0	4	5	6	8	12	12	18	19
9	d	0	4	5	6	8	12	12	18	19

Key relaxations:

$$d(g) = d(h) + 1 = 6,$$

$$d(f) = d(g) + 2 = 8,$$

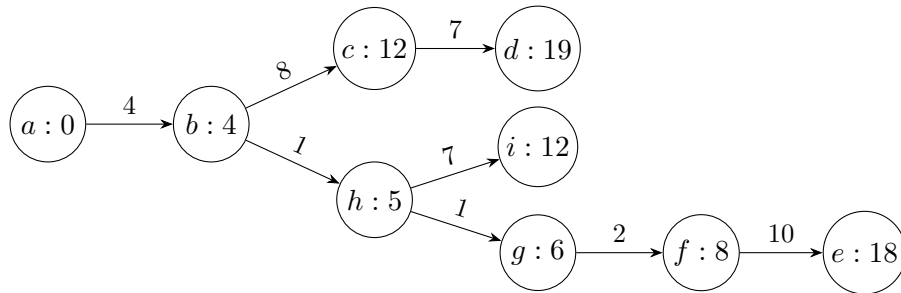
$$d(d) = d(f) + 14 = 22 \quad \text{initially,}$$

$$d(d) = \min(22, d(c) + 7) = \min(22, 19) = 19.$$

$$d(i) = d(h) + 7 = 12,$$

$$d(e) = d(f) + 10 = 18,$$

Final shortest-path tree:



Node labels: vertex and distance.

AStar1 (p. 8)

$f(n) = g(n) + h(n)$; g = best known cost from A .

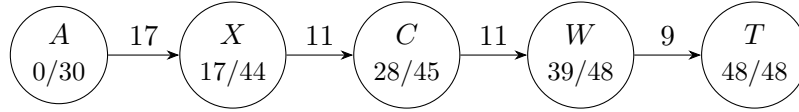
n	A	X	B	C	L	W	K	T	S	H
$h(n)$	30	27	16	17	33	9	19	0	17	22

OPEN = $\{A(0, 30)\}$. Notation $n(g, f)$; OPEN sorted by f .

Step	Pop	Calculations	OPEN after step
1	A	$g(X) = 0 + 17 = 17, f(X) = 17 + 27 = 44$	$X(17, 44)$
2	X	$g(B) = 17 + 16 = 33, f(B) = 49$; $g(C) = 17 + 11 = 28, f(C) = 45$	$C(28, 45), B(33, 49)$
3	C	$g(W) = 28 + 11 = 39, f(W) = 48$; $g(L) = 28 + 20 = 48, f(L) = 81$	$W(39, 48), B(33, 49),$ $L(48, 81)$
4	W	$g(T) = 39 + 9 = 48, f(T) = 48$; $g(K) = 39 + 10 = 49, f(K) = 68$; $g(B) : 51 > 33$ (reject).	$T(48, 48), B(33, 49),$ $K(49, 68), L(48, 81)$
5	T	Goal popped; stop.	$B(33, 49), K(49, 68),$ $L(48, 81)$

CLOSED : A, X, C, W, T .

$$\boxed{A \rightarrow X \rightarrow C \rightarrow W \rightarrow T}, \quad \boxed{17 + 11 + 11 + 9 = 48}.$$



Node labels: g/f . Stop when the goal is popped.

AStar2 (p. 9)

$h(B) = 6$; relax and update OPEN entries and parents.

Step	Pop	Calculations	OPEN after step
1	A	$g(X) = 17, f(X) = 44$	$X(17, 44)$
2	X	$g(B) = 33, f(B) = 33 + 6 = 39;$ $g(C) = 28, f(C) = 45$	$B(33, 39), C(28, 45)$
3	B	$g(W) = 33 + 12 = 45, f(W) = 54;$ $g(T) = 33 + 17 = 50, f(T) = 50;$ $g(S) = 33 + 7 = 40, f(S) = 57$	$C(28, 45), T(50, 50),$ $W(45, 54), S(40, 57)$
4	C	$g(W) : 45 \rightarrow 39, f(W) = 48, p(W) = C;$ $g(L) = 48, f(L) = 81.$	$W(39, 48), T(50, 50),$ $S(40, 57), L(48, 81)$
5	W	$g(T) : 50 \rightarrow 48, p(T) = W; g(K) = 49,$ $f(K) = 68; g(B) : 51 > 33 \text{ (reject).}$	$T(48, 48), S(40, 57),$ $K(49, 68), L(48, 81)$
6	T	Goal popped; stop.	$S(40, 57), K(49, 68),$ $L(48, 81)$

Result:

$$\boxed{A \rightarrow X \rightarrow C \rightarrow W \rightarrow T \text{ with cost } 48}.$$

Expansion order: A, X, B, C, W, T .

Without OPEN relaxation: $W(45, 54)$ stays stale; $T(50, 50)$ is popped, giving the wrong path $A \rightarrow X \rightarrow B \rightarrow T$ (cost $50 > 48$).

$h(B) \in \{6, 16\} \leq 17 = h^*(B)$: admissible. With $h(B) = 6$, consistency fails:

$$h(X) = 27 > 16 + h(B) = 22, \quad h(S) = 17 > 7 + h(B) = 13.$$

Inconsistent heuristic: reopen CLOSED if a better g appears. This run needs only OPEN updates.

AStar3 (p. 10)

(a) Continuous positions: $\mathcal{S} \subset \mathbb{R}^2$, $|\mathcal{S}| = \infty$ (uncountable).

Visibility graph: vertices S, G and obstacle corners; edges are unobstructed straight segments.

$$5 + 3 + 4 + 4 + 4 + 3 + 6 = 29 \text{ vertices.}$$

$|V_{\text{visibility}}| = 29 + 2 = 31$ (reduced graph, not all physical positions).

(b) Triangle inequality: $|AC| \leq |AB| + |BC|$ (strict for a non-collinear bend). Straighten free-space bends; essential bends occur at obstacle vertices.

Shortest path: straight segments through visible corners; some segments follow boundary edges.

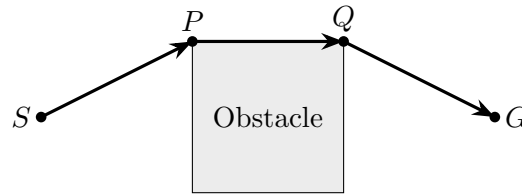


Illustration of the geometric principle, not the original scene

Assume a point robot, Euclidean distance, boundary contact allowed.